Review of "Koopman Spectral Theory for Indivisible Stochastic Processes"

1 Overall assessment

This manuscript develops a coherent operator-theoretic framework for analysing two-time stochastic transition families that fail the Chapman-Kolmogorov composition property. The central use of the deficit operator provides a clear organising principle, linking the global input and output deficit subspaces with a local second-order measure of indivisibility. In particular, the local expansion involving $\beta(s)$ is a potentially useful diagnostic, while the approximation bound gives a transparent quantitative interpretation of the deficit. The manuscript also handles several important distinctions carefully, including those between Markovianity and two-time divisibility, forward transition and Koopman operators, and intrinsic stochastic information and gauge-dependent unitary representations. The worked examples are helpful in illustrating the abstract constructions. The attempt to connect the deficit framework with continuous Stinespring dilations and causal trajectory-space operators is ambitious and potentially valuable. Overall, the paper contains interesting ideas and could make a useful contribution following resolution of the mathematical and interpretive issues detailed below.

2 Major comments

The introduction would benefit from a more theorem-level comparison with existing approaches to non-Markovian transfer operators, cocycle defects and process-tensor formulations. At present, the precise novelty of the deficit-based construction relative to these literatures is not sufficiently transparent.

Since the primary objects are forward transition operators and singular subspaces of their composition deficits, the author should clarify why the framework constitutes "Koopman spectral theory" rather than an operator-theoretic analysis of Chapman-Kolmogorov defects.

The same claimed contributions are repeated in the abstract, introduction, relationship-to-prior-work discussion, outline and later discussion. This repetition obscures the manuscript's principal mathematical contributions. The abstract alone mentions almost every theorem, condition, caveat, example and open problem, and could be shortened.

The relationship between Markovianity and two-time Chapman-Kolmogorov divisibility is standard rather than a novel contribution. The manuscript should present Proposition 2.6 and Corollary 2.7 explicitly as background results and avoid including this hierarchy among the paper's stated original contributions.

The manuscript permits general two-time transition families that need not be induced by a consistent stochastic process. This operator-level generality is legitimate, but the empirical interpretation requires greater qualification. A nonzero estimated deficit may reflect genuine memory, omitted state variables, nonstationarity, sampling error, inconsistent kernel estimation, or even the absence of a joint process compatible with the fitted pairwise kernels. The author

should distinguish these possibilities and avoid presenting $\Delta \neq 0$ alone as evidence of intrinsic physical indivisibility.

The claim that Bell-type arguments exclude any measurable divisible or Markovian augmentation appears too broad. A stochastic process may generally be Markovised by enlarging the state to include its history. The author should state precisely what restrictions are imposed on the augmentation-such as locality, dimension, observable structure or physical admissibility-and cite a theorem supporting the resulting claim.

In Proposition 2.11(c), the matrix $\frac{1}{2}(J - I)$ is doubly stochastic only for $n = 3$; for general n, its row and column sums equal $(n - 1)/2$. The author should specify that this is the 3×3 example and explain how strictness is extended to $n > 3$.

The contraction assumption is clearly stated, but the manuscript should consistently qualify the later $1/3$ bound as universal only within the contraction class. Column-stochasticity alone does not imply $L^2(\mu)$-contractivity.

Section 3.2 should distinguish a time-homogeneous C_0-semigroup from a non-autonomous divisible evolution family. Divisibility, strong continuity and contractivity alone do not make $P(t, s)$ a semigroup in elapsed time.

Proposition 3.6 correctly excludes representation by a time-ordered exponential of a bounded Bochner-integrable generator. The broader wording in the abstract and introduction-suggesting that no generator "of any kind" exists-should be restricted to the class covered by the proposition. Unbounded evolution generators, memory-kernel equations and enlarged-state representations are not excluded.

The input and output deficit subspaces provide a useful description of where composition failure acts. However, much of Theorem 4.5(a) follows immediately from defining V_{in} and V_{out} as the closed spans of the ranges of Δ^* and Δ, respectively, together with standard compact-operator theory. The manuscript should calibrate its novelty claims accordingly and explain what new dynamical information these subspaces provide beyond this structural repackaging.

The statement following Corollary 4.6 that the lower bound $D(t, s, r)/3$ is "invariant under the choice of intermediate time r" appears misleading, since $D(t, s, r)$ generally depends on r. Presumably the intended meaning is that the inequality holds for every choice of r. This should be corrected.

The empirical protocol also requires stability analysis for V_{in}, V_{out} and their leading singular directions under kernel-estimation error, particularly when singular values are clustered or close to zero.

The discussion should qualify the interpretation of Corollary 4.6. The result obstructs uniform operator-norm approximation across the three time pairs of a selected triple under the contraction assumptions; it does not imply that every observable, marginal quantity or individual transition pair is separated from all divisible approximations by $D/3$.

In Appendix C, the proposed application of Michael's continuous-selection theorem is not valid as stated: the set of orthonormal frames of a subspace is not convex. The continuous-completion result may instead follow from the vector-bundle triviality argument over the contractible parameter domain, but the proof should rely on a correct theorem and provide the necessary relative trivialisation along the diagonal.

The proof of Theorem 5.4(ii) contains a regularity gap. The Stinespring isometry in (18) depends on $\sqrt{\Gamma_{ij}(t \leftarrow s)}$, whereas (R3′) assumes absolute continuity only of Γ_{ij}. Absolute continuity and strict positivity for $t > s$ do not in general imply absolute continuity of the square root, particularly when an off-diagonal entry approaches zero as $t \downarrow s$. Consequently, the assertion in

Appendix C that $V(t,s)$ is absolutely continuous under (R3′) is not justified, and the subsequent construction of an L^1 Hamiltonian does not follow as stated. The theorem could be repaired by assuming directly that $t \mapsto \sqrt{\Gamma_{ij}(t \leftarrow s)}$ is absolutely continuous with integrable derivative, or by imposing sufficiently strong quantitative regularity near zeros.

The finite-grid cocycle in Theorem 5.4(iii) is obtained by assigning a fresh environment to each increment and defining the multi-step operators as ordered products. The resulting cocycle identity is therefore automatic. The manuscript should distinguish this standard repeated-interaction construction from the unresolved and substantially stronger problem of a fixed-environment continuum cocycle.

In Appendix C, minimality is a pointwise property of the dilation of $\Phi(t,s)$. Thus zero Kraus operators should be removed at the particular time pair under consideration, not only when the corresponding transition entry vanishes identically over the entire parameter domain.

The manuscript should clarify the compatibility of its standing assumptions. On an infinite-dimensional $L^2(\Omega,\mu)$, compactness of $P(t,s)$ for every $t > s$, together with operator-norm continuity as $t \downarrow s$ and $P(s,s) = I$, would imply that the identity is compact, which is impossible. Thus the compact singular-value framework of Section 4 and the norm-continuous trajectory-space framework of Section 7 do not apply simultaneously in the stated infinite-dimensional generality. The domains of applicability of the different results should be separated explicitly.